

Project Information

Research Project

Reasoning Under Uncertainty · 605.745

Overview

This course project asks you to apply a formal reasoning-under-uncertainty framework to a real decision or inference problem - either by critically applying an existing software package, or by implementing the reasoning method yourself. At this level, a project should go beyond a worked textbook example: it should make an explicit modeling choice, evaluate that choice quantitatively, and honestly discuss where the method breaks down.

The project must include an introduction describing the problem domain and why the reasoning framework you chose fits that domain.

Learning Objectives

After completing this project, you will be able to:

- Formulate a real-world problem as a reasoning-under-uncertainty task, and justify your choice of formalism against at least one plausible alternative.
- Build or configure a working implementation of that formalism and validate it against test cases with known or expected outcomes.
- Evaluate your approach quantitatively - accuracy, calibration, robustness to noisy or missing evidence, or computational cost - rather than relying on descriptive claims alone.
- Critically analyze the limitations of your chosen technique, and state what would need to change for it to generalize beyond your project's example.
- Communicate your work in a paper and presentation aimed at a technical, non-classroom audience.

Getting Started & the Project Prompt

Select (or propose, subject to instructor approval) a decision-making or inference problem that is genuinely uncertain: the evidence is noisy, incomplete, conflicting, or probabilistic by nature.

Representative domains include medical or fault diagnosis from imperfect sensors, autonomous-agent decision-making under partial observability, fraud or anomaly detection from conflicting signals, forecasting from sparse data, and multi-sensor fusion. Domain familiarity is welcome, but the *reasoning framework*, not the domain, is what's being evaluated.

The first step is to select a subject from the list provided in the Course Project folder, or contact me to propose your own. The second step is to follow the Project Checklist in the Course Project folder so you know all deliverables and what they entail.

Required Elements

Every project, regardless of topic, must include all four of the following. These map directly onto the rubrics below.

- A **named uncertainty-reasoning framework**, applied through an existing package or implemented directly, with a stated justification for why it fits the problem better than at least one named alternative.
- A **working artifact**: runnable code, or — if a full implementation is infeasible — detailed, executable-level pseudocode plus a worked numerical trace.
- A **quantitative evaluation** reporting at least one metric (e.g., classification accuracy, log-loss or Brier score, mean absolute error, posterior calibration, runtime vs. problem size) computed across more than a single instance.
- A **comparison point**: a second method, a baseline (e.g., a naive rule-based classifier, a simpler probabilistic model), or a sensitivity analysis (varying priors, evidence quality, or structure) that lets the reader judge whether the added complexity was worth it.

Explicitly Out of Scope

A literature survey with no implementation, an unmodified walkthrough of a library's documentation example, or a purely qualitative discussion with no reported metric will not satisfy the Research Methods criterion below, regardless of writing quality.

Deliverables & Grading Rubrics

Each rubric below uses the same three-band structure (**Emerging / Developing / Proficient**) so you can see exactly what separates one band from the next on every deliverable. Specific due dates are posted on the course calendar.

1. Project Proposal

A one-page proposal: the problem, why it is genuinely uncertain, the candidate reasoning framework and the alternative it is being chosen over, and how success will be measured.

Rubric: Project Proposal

Criterion	Weight	Emerging	Developing	Proficient
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Criterion	Weight	Emerging	Developing	Proficient
Problem Definition & Relevance	25%	Problem described in general terms; not clear the outcome is actually uncertain.	Concrete and plausibly uncertain, but the source of uncertainty is not named.	Concrete, and names the specific source of uncertainty (noise, missing/conflicting evidence).
Framework Selection & Justification	30%	Names a framework with no reason for the choice, or a poor fit for the problem.	Names a framework and a reason, but no named alternative for comparison.	Names a framework, a specific alternative, and a concrete reason for preferring one.
Comparison / Baseline Plan	20%	No comparison or baseline proposed.	A comparison is mentioned but no method is named.	A specific baseline or sensitivity analysis is named and briefly described.
Evaluation Plan	25%	No metric proposed; success defined only qualitatively.	A metric is named but data/test cases are unspecified.	A specific metric is named along with the data or test cases it will be computed on.

2. Annotated Outline

A section-by-section outline of the final paper with a short annotation under each heading describing what will go there, plus an initial reference list.

Rubric: Annotated Outline

Criterion	Weight	Emerging	Developing	Proficient
Structural Completeness	30%	Missing two or more required sections (Overview, Problem Statement, Study Approach, Results, Conclusion, References).	Missing one required section, or sections are out of coherent order.	All required sections present and logically ordered.
Literature Foundation	30%	Fewer than 3 sources, or non-technical sources with no annotation.	3–4 credible technical/peer-reviewed sources, each with a one-sentence relevance note.	5+ credible sources, each annotated with how it will be used (background, method, comparison, or data).

Criterion	Weight	Emerging	Developing	Proficient
Section-Level Clarity	20%	Headings only, no description of planned content.	Most sections have a one-sentence description.	Every section has a 2–4 sentence description specific enough to predict its content.
Alignment with Proposal	20%	Contradicts or omits what was proposed in Deliverable 1.	Broadly consistent, but hasn't incorporated feedback given on the proposal.	Reflects the approved framework, comparison plan, and any instructor feedback.

3. Rough Draft of Operable Code

A working prototype (or, where a full implementation is infeasible, detailed executable-level pseudocode) implementing the core reasoning method, plus at least two worked test cases.

Rubric: Rough Draft of Operable Code

Criterion	Weight	Emerging	Developing	Proficient
Functional Correctness	35%	Code doesn't run, or pseudocode has gaps that would prevent execution.	Runs, but only on the simplest case, or with unexplained errors on realistic input.	Runs end-to-end (or pseudocode is complete and hand-traced) with output consistent with expectations.
Implementation of the Reasoning Framework	30%	The reasoning step is a black-box library call with no evidence you understand what it computes.	Implemented/configured correctly; shows at least one intermediate result (posterior, belief mass, value function) with an explanation.	As Developing, plus demonstrates awareness of at least one modeling assumption built into the framework.
Test Cases & Validation	20%	No test cases, or a single untraceable example.	At least 2 test cases with stated inputs and observed outputs.	At least 2 test cases with independently derived expected outputs, compared against observed results.

Criterion	Weight	Emerging	Developing	Proficient
Documentation & Reproducibility	15%	No run instructions and no in-line comments.	Some comments or a partial run instruction; a third party would still have questions.	Clear run instructions plus comments sufficient for a third party to reproduce the result unaided.

4. Rough Draft of Paper

A complete draft of every section listed under Deliverable 5, understood as a draft: results may be preliminary, but every section must exist and make an argument, not hold a placeholder.

Rubric: Rough Draft of Paper

Criterion	Weight	Emerging	Developing	Proficient
Completeness of Sections	25%	Two or more required sections missing or placeholder text.	All sections present; one is materially underdeveloped.	All sections present and developed enough for substantive feedback.
Coherence of Argument	25%	Doesn't state what question it's answering, or sections don't build toward a conclusion.	A thesis/question is stated but not clearly connected to the results.	A thesis/question is stated in the introduction, and results/conclusion clearly speak back to it.
Preliminary Results Reported	25%	No numbers or output reported; results described only in words.	At least one numeric result reported, without a baseline or comparison.	At least one numeric result reported alongside the comparison/baseline named in Deliverable 1.
Clarity for Feedback	25%	Difficult to follow; a reviewer can't	Generally clear with some rough passages or unsupported claims.	Clear enough that feedback can target specific claims and sentences.

Criterion	Weight	Emerging	Developing	Proficient
		identify what to comment on.		

5. Case Study Paper (Final)

Required sections: Title Page, Overview, Problem Statement, Study Approach, Results, Conclusion, References, Figures, Appendices (as applicable).

Rubric: Case Study Paper

Criterion	Weight	Emerging	Developing	Proficient
Content & Thesis	20%	No clear research question, or the paper drifts from it after the introduction.	A research question is stated and mostly maintained, but at least one section doesn't advance it.	A clear, specific research question is stated and every section visibly advances or answers it.
Organization & Structure	15%	Missing required sections, or section order obscures the argument.	All required sections present; transitions between at least two sections are abrupt.	All required sections present, logically ordered, with transitions that carry the argument forward.
Critical Analysis & Depth	20%	Results are reported without discussion of why they occurred or where the method could fail.	Discusses at least one limitation, but doesn't connect it to the specific results obtained.	Connects specific limitations to specific results, and considers at least one counter-explanation.
Research Methods & Rigor	25%	Fewer than 5 credible sources cited, or no comparison/metric reported.	5+ sources cited and referenced; comparison and metric present but not discussed in the conclusion.	5+ sources cited and referenced; a named comparison and metric are present, and the result is explicitly discussed in the conclusion.

Criterion	Weight	Emerging	Developing	Proficient
Results & Quantitative Evaluation	20%	Results are qualitative only, or the metric is computed on a single instance with no indication of variability.	A metric is reported across multiple trials, but with no measure of spread, confidence, or sensitivity.	A metric is reported across multiple trials with a measure of spread/confidence or a sensitivity analysis supporting the conclusion.

6. Presentation

A 10–12 minute presentation (PowerPoint/PDF or recorded demonstration) of your method and findings.

Rubric: Presentation

Criterion	Weight	Emerging	Developing	Proficient
Time Requirement	25%	Under 8 minutes or over 15 minutes.	8–10 minutes or 13–15 minutes.	10–12 minutes.
Content	25%	Restates the proposal with no results, or omits the comparison/baseline entirely.	Covers method and results, but the comparison's outcome isn't stated.	Covers method, results, and the comparison's outcome in a way a non-specialist can follow.
Delivery	25%	Starts late or is visibly unrehearsed.	On time, generally well-paced, occasional reliance on notes.	On time, well-paced throughout, notes used only as a light reference.
Comprehension (Q&A)	25%	Can't answer a direct question about the method or the comparison.	Answers direct questions correctly but with visible hesitation.	Answers questions about method, comparison, and limitations accurately and without hesitation.

Submission

The project has incremental deadlines through the term so you get feedback early enough to act on it, see the course calendar for exact dates. The **Case Study Paper is due the Monday of Module**

14, with presentations on the last day of class.

Going Further

Projects that include a rigorous quantitative comparison, a solid literature grounding, and honest discussion of limitations aren't just stronger on this rubric, they're also the ones most likely to be worth extending into a workshop paper, poster, or portfolio piece after the course ends. If you think your project has that potential, come talk to me before the final draft is due.

Plagiarism & Academic Integrity

Plagiarism is taking the words, ideas, or code of another and presenting them as your own. If you use another's ideas, cite the source completely. If you use another's words or code, mark them clearly (quotation marks, indentation, or code comments crediting the source) and cite completely.

This applies equally to text or code generated with AI assistance: **any AI-assisted writing or coding must be disclosed** in the paper's acknowledgments, consistent with course and university policy.